



EARTH SCIENCE

Layers of the Atmosphere:

1. Troposphere

- Also called lower atmosphere; clouds, rainfall and other weather conditions occur ONLY here.

2. Stratosphere

- Contains 80% of the atmosphere's mass; contains 99% of air along with troposphere.

3. Mesosphere

- Temperature decreases with increasing height; upper layer is MESOPAUSE the coldest naturally occurring place on earth.

4. Thermosphere

- Temperature increases with increasing height; biggest of all the layers.

5. Exosphere

- Gas molecules achieve ESCAPE VELOCITY until they move out to outer space.

SPECIAL LAYERS:

1. Ionosphere

- Composed of the exosphere, thermosphere and parts of mesosphere; IONIZED by the sun's rays.
- Helps in propagation of RADIO WAVES

2. Ozone

- Serves to protect the earth from harmful rays of the sun.
- Highest concentration in the STRATOSPHERE

METEOROLOGY: study of the atmosphere and factors affecting it.

	WEATHER	CLIMATE
Definition	General condition of the atmosphere	Average condition of the atmosphere
Scope	Specific area	Extended region
Period of Time	Brief (within 24 hours)	Long (not less than 1 year)

FACTORS AFFECTING WEATHER AND CLIMATE:

- Air pressure** – the weight/ force of air on a unit area; measured by BAROMETER; the denser and cooler the air, the greater the air pressure.
- Humidity** – amount of water vapor in the air; measured by HYGROMETER; air containing much water vapor is said to be HUMID.

- Air mass** – large body of air which extends over thousands of kilometers and has the same temperature, pressure and humidity at all points. The surface of contact between two air masses is called an AIR FRONT.

CLOUDS: formed when evaporated water rises above the atmosphere and then condenses as it cools.

- Cumulus** – piles of puffed cotton which indicates fair weather.
- Stratus** – gray in appearance; low lying, formed from humid air.
- Cirrus** – white and feathery streaks, high lying, formed from low humidity areas.
- Nimbus** – thick, dark and rain producing

PRECIPITATION: process by which condensed water vapor falls from the clouds to the ground.

- Rain** – liquid form of water
- Snow** – ice crystals
- Sleet** – mixed rain and snow
- Hail** – ice crystals formed with various layers and size ranging from that of mongo seed to a man's fist.

MONSOONS: Seasonal changes in atmospheric circulation and precipitation due to asymmetric heating of land and sea.

- Sea breeze** – during the day, air over the land is warmer than over the sea. Cool dense air from the sea moves toward the land as a sea breeze and pushes the warm air upwards.
- Land breeze** – during the night, air over the sea is warmer than over the land. Cool dense air from the land moves toward the sea as a land breeze and pushes the warm air upwards.
- Northeast monsoon (Amihan)** – blows from November to April causing rainy weather
- Soutwest monsoon (Habagat)** – blows from May to October causing rainy weather.

TROPICAL CYCLONES

- Low pressure center over an ocean with strong circulating winds.
- EYE:** center of a cyclone where warm air rises.



TYPES:

1. **Depression:** maximum winds of less than 60 km/ hr
2. **Storm:** maximum winds of 60-118 km/ hr
3. **Typhoon:** maximum winds of 119 km/ hr or above

GREENHOUSE EFFECT: THE PROCESS

- About 50% of the sun's energy is absorbed at the Earth's surface and the rest is reflected by the atmosphere (largely by CLOUDS).
- The absorbed energy warms the surface of the Earth, to a temperature around 255 K, radiated infrared heat.
- The heat is absorbed in each layer of the atmosphere through the GREENHOUSE GASES.
- To balance temperature, the atmosphere re-radiates the absorbed heat, both upwards (to the outer space) and downwards (to the Earth).

THE EARTH'S LITHOSPHERE – rigid solid shell of the earth.

THEORIES:

1. PANGAEA

- The single major continent which broke into two pieces (LAURASIA and GONDWANALAND). These pieces further broke apart and drifted away to form the continents of today as stated by Alfred Wegener in the CONTINENTAL DRIFT THEORY.

2. ISOSTASY

- The earth's crust tends to distribute itself over the mantle to maintain a state of balance. (1) the crust floats over the mantle and (2) high density and massive parts of the crust sink while low density and thinner parts rise.

3. MILD-OCEANIC RIFT

- Remains of continents that parted and formed ridges from which magma flows out.
- **LAVA** – magma that reaches the earth's surface.
- **Seafloor spreading** – caused by magma continuously pouring out of the mid-oceanic rift, hardening and forming new ocean floor.

4. PLATE TECTONIC THEORY

- The earth is made up of several plates which move on top of the mantle. Collision of plates result to either the edge of the less dense plate crumples and forms MOUNTAINS or one plate sinks below the other and forms a TRENCH.

LAYERS OF THE EARTH

1. CRUST

- 35 kilometers thick and is composed of sedimentary, igneous and metamorphic rocks.

- 35% CONTINENTAL CRUST, which has more silicon, potassium but less iron, magnesium and calcium than the OCEANIC CRUST.
- **Mohorovic discontinuity** – layer between the crust and the mantle where the tectonic plates move.

2. MANTLE

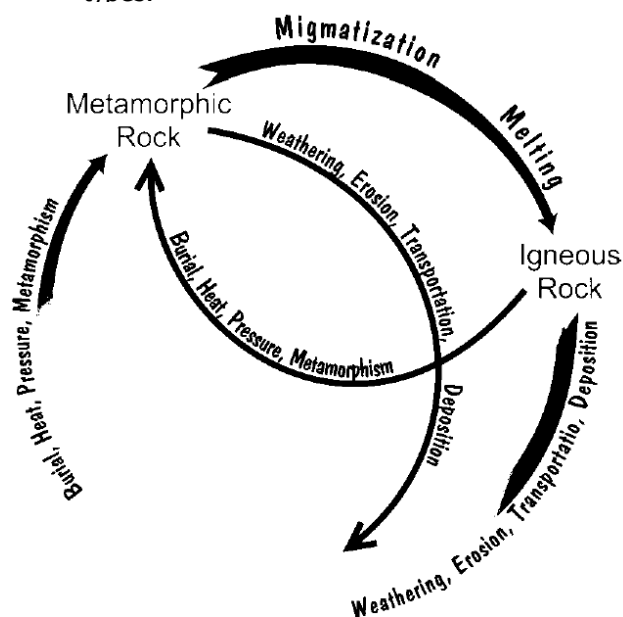
- Second layer of the earth and includes most of the earth's volume and mass. About 20 km to 2,900 km thick and denser than crust.
 - a. **Upper mantle:** partially molten
 - b. **Lower mantle:** solid
 - c. **Gutenberg discontinuity:** layer between the Earth's mantle and core

3. CORE

- Center most layer, with radius of 3,500 km thick and densest of all layers.
 - a. **Outer core:** liquid
 - b. **Inner core:** solid composed of Iron, Nickel, Silicon

ROCK CYCLE: process by which rocks change in kind; rocks and soil are the solid material in earth.

1. **Sedimentary** – formed by deposition, accumulation, cementing and hardening of loose sediments into rock.
2. **Igneous** – formed by the cooling of magma. Most common type.
3. **Metamorphic** – formed when igneous and sedimentary rocks are changed by increased pressure and temperature. Hardest of the three types.



MINERALS

- Naturally occurring substances with a definite, geometric arrangement of atoms and molecules which makes up rocks.
- Nonrenewable resources.



Mohs' Hardness Scale: measures the ability of a mineral to resist scratch.

- ✓ **TIP:** The higher the scale, the harder the mineral

Scale	Mineral	Simple Hardness
10	Diamond	None
9	Ruby, Sapphire, Alexandrite, Chrysoberyl	None
8	Aquamarine, Emerald, Morganite, Spinel, Topaz	None
7	Amethyst, Citrine, Carnet, Ioite, Tourmaline, Zircon	Scratches window glass
6	Diopside, Lapis lazuli, Orthoclase, Turquoise	Can be scratched with steel file
5	Apatite, Obsidian, Sphene, Glass	Can be scratched with a knife
4	Coral, Fluorite, Pearl	Can be easily scratched with a copper coin
3	Calcite, Ivory	Can be scratched with a fingernail
2	Amber, Gypsum	Can be scratched with a fingernail
1	Talc	Can be scratched with a fingernail

HYDROSPHERE – combined mass of water on earth
FEATURES:

- 1. WAVES AND CURRENTS** – movement of water by force of the wind
- 2. TSUNAMIS** – movement of water by earthquakes
- 3. TIDES** – movement of water caused by the gravitational pull of the sun and moon on earth

WATER FACTS:

- ✓ 71% of the earth's surface is covered with water
- ✓ Water is the universal solvent due to its POLAR structure (has positive and negative ends)
- ✓ Water gives off and absorbs large amounts of heat
 - Melting of ice and evaporation – absorbs heat (ENDOTHERMIC)
 - Freezing and Condensation – releases heat (EXOTHERMIC)
- ✓ Salts in seawater give it GREATER DENSITY and LOWER FREEZING POINT than pure water

ASTRONOMY

- Sun and the eight planets revolving around it: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune
- Pluto was now removed from the list (dwarf planet)
- **Nebular hypothesis** – most widely accepted theory on the origin of the solar system. States that the solar system was formed from a huge

NEBULA which contracted, rotated and condensed into hot lumps of matter to form the sun and planets.

EARTH	1. Shape	Oblate spheroid
	2. Rotation	The movement of the earth on its own axis in a west to east direction in a 24 hour cycle
	3. Revolution	The movement of the earth around the sun from west to east in 365 ¼ days. Seasons are caused by the changing orientation of the earth's axis with respect to the sun as the earth revolves.
	4. Tilt of axis	23 ½ degrees, thus the lengths of daytime and nighttime continually change but always add up to 24 hours.
OTHER PLANETS	1. Ringed planets	Saturn, Jupiter, Uranus and Neptune
	2. Terrestrial planets	Mercury, Venus, Earth and Mars

MOON

- ✓ The earth's satellite is the moon
- ✓ The moon has no atmosphere
- ✓ Doesn't have its own light but reflects the light of the sun.

ECLIPSES:

- 1. Lunar eclipse** – occurs when the earth is between the moon and the sun. The earth's shadow darkens the moon.
- 2. Solar eclipse** – occurs when the moon is between the earth and the sun. The moon's shadow casts over the earth.

THE UNIVERSE

- **Big Bang Model** – The universe was once in an extremely hot and dense state that expanded rapidly. This rapid expansion caused the young Universe to cool resulting to its present CONTINUOUSLY EXPANDING state.

STARS: Luminous ball of plasma composed of 73% Hydrogen and 25% Helium

- **Red star** – coolest
- **White blue star** – hottest

GALAXY: A massive, gravitationally bound system that consists of stars and stellar remnants. The sun and the earth is situated at the MILKY WAY GALAXY.

SATELLITE: an object that orbits another object usually a planet.